

Assignment 1: Pattern Search

By using Qiskit/Python and QuantumSavory/Julia, you are tasked with solving a pattern search problem using quantum computing. The goal is to design and implement a quantum algorithm capable of identifying specific patterns in a 3×3 grid of binary pixels. Each pixel can be either "on" (1) or "off" (0), forming a total of $2^9 = 512$ possible images. An oracle classifies images as "good" or "bad" based on hidden criteria.

Problem Statement

Design and implement on Qiskit/Python and QuantumSavory/Julia a quantum algorithm that solves the following problem: Given a black-box oracle, identify at least one image that is classified as "good" by the oracle.

Details and Objectives

1. Oracle Description

- The oracle implements a function $f : \{0, 1\}^9 \rightarrow \{0, 1\}$ where $f(x) = 1$ for "good" images and $f(x) = 0$ for "bad" images.
- The oracle's definition is hidden, and your algorithm must work as a general solution without knowledge of the specific classification rules.

2. Classical vs Quantum

Compare the efficiency and performance of your quantum algorithm with a brute-force classical search for the same task. Evaluate:

- The number of oracle calls required.
- Execution time on noiseless simulators and real quantum hardware. Use bit flip and depolarizing quantum noise models.

3. Evaluation on Example Oracles

Implement at least two example oracles with the following criteria:

- The pattern contains a vertical line of three "on" pixels.
- The pattern contains a 4-"on"-pixel square (::).
- Optionally, create additional custom oracles with more complex patterns.

4. Performance Metrics

Measure and compare:

- Algorithm performance on noiseless simulators (using AerSimulator).
- Algorithm performance on real quantum hardware (using IBM Quantum devices or Fake providers).

5. Project Presentation

During the project presentation, the candidates, in addition to illustrating the results obtained, must show the quantum circuit that implements the oracle.